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Lauren Hoffman

Research Interests

Polar Science, Sea Ice, Physical Oceanography, Air–Sea–Ice Interactions, Climate Variability and Predictability, Machine Learning, Explainable Artificial Intelligence, AI-Enabled Observing System Design.

Education

09/2023 **Ph.D., Chemical Engineering**, *University of California San Diego*, San Diego, CA

06/2019 **M.S., Chemical Engineering**, *University of California San Diego*, San Diego, CA

05/2016 **B.A./B.S., Mechanical Engineering**, *University of San Diego*, San Diego, CA
Mathematics and Chemistry minors; Honors Program. Dean's List, First Honors.

Research Experience

01/2026–Present **Postdoctoral Researcher**, *University of California Santa Cruz*, Santa Cruz, CA, Advisor: Xavier Prochaska
- Characterizing density-based ocean fronts and associated submesoscale dynamics in high-resolution ocean model output.
- Developing scalable analysis and visualization workflows for large ocean model datasets.

02/2024–12/2025 **Postdoctoral Researcher**, *Université catholique de Louvain*, Louvain-la-Neuve, Belgium, Advisor: François Massonnet
- Developed machine learning and explainable AI approaches to predict and understand rapid Arctic sea ice loss events on seasonal to interannual timescales.

10/2023–01/2024 **Postdoctoral Researcher**, *Scripps Institution of Oceanography, UC San Diego*, San Diego, CA, Advisor: Matt Mazloff
- Implemented components of an operational forecast system for sea-ice motion prediction using machine learning.

06/2019–09/2023 **Graduate Researcher**, *Scripps Institution of Oceanography, UC San Diego*, San Diego, CA, Advisor: Matt Mazloff
- Investigated upper-ocean salinity response to atmospheric river events in the California Current System using observations and ocean model output.
- Developed machine learning and explainable AI approaches to predict and understand Arctic sea-ice dynamics from remote sensing data.

09/2018–03/2019 **Graduate Researcher**, *University of California San Diego*, San Diego, CA, Advisor: Andrea Tao

03/2018–08/2018 **Research Assistant**, *University of San Diego*, San Diego, CA, Advisor: Truc Ngo

03/2015–06/2017 **Undergraduate Research Assistant and Honors Thesis**, *University of San Diego*, San Diego, CA, Advisor: Truc Ngo

Teaching and Mentoring

2020 & 2021 **Instructor**; Climate Change and the Ocean; UC San Diego Academic Connections; 4-week online pre-college course; students accessed, plotted, analyzed and presented real oceanographic data

09/2018–12/2020 **Teaching Assistant**; Chemical Engineering, UC San Diego; fluid mechanics (8 offerings), material and energy balances (1); 5 hrs/week of lecture and discussion sections

2025 **Guest Lecturer**; Introduction to Machine Learning for Earth and Climate Science; Forecast, Prediction, and Projection in Climate Science; UCLouvain

2023 **Guest Lecturer**; Introduction to Machine Learning for Earth and Climate Science; Analysis of Physical Oceanographic Data; Scripps Institution of Oceanography

2024–2025 **Co-Mentor**; Antonio Martinez Soares, Master's Student; Université catholique de Louvain

2021 **Co-Mentor**; Kayli Matsuyoshi, Undergraduate Student; Scripps Institution of Oceanography

2020–2021 **Co-Mentor**; Aniruddh Varadarajan, Undergraduate Student; University of California San Diego

Leadership and Professional Service

2026 **Invited Participant**; Roundtable on climate modelling with the UN Secretary-General's Scientific Advisory Board

2025–Present **Co-Lead**; OAISIS Working Group; international Antarctica InSync working group on AI-enabled approaches to Antarctic sea ice observing-system design and recommendations

2025–Present **Working Group Member**; ORCAS (Observational Requirements in the Context of AI prediction Systems)

2025 **Student Poster Judge**; AMS Conference on Polar Meteorology and Oceanography

2022–2025 **Mentor Group Member**; MPOWIR (Mentoring Physical Oceanography Women+ to Increase Retention)

2021 **Pod Member**; URGE (Unlearning Racism in the Geosciences)

2013–2015 **Fundraising Director**; Engineers Without Borders

Peer Review Journal of Climate; Geophysical Research Letters; Journal of Geophysical Research: Atmospheres; npj; Environmental Modelling & Software; Open Science Europe.

Honors and Awards

2021 Interdisciplinary Research Award, UC San Diego.

2020 Teaching Assistant Commendation, "For Extraordinary Impact as a Teaching Assistant."

2015–2016 Achievement Rewards for College Scientists (ARCS), $5,000.

2011–2016 Alcala Award, Merit Based Scholarship, $20,000 annually.

2015–Present Tau Beta Pi, Engineering Honor Society.

2016–Present Pi Tau Sigma, Engineering Honor Society.

Technical Expertise

Programming Python; MATLAB; UNIX/Linux; LaTeX; BibTeX.

Software/Tools TensorFlow/Keras; Git; Claude.

Methods Machine learning; neural networks; explainable artificial intelligence; predictive modeling; data analysis; remote sensing; numerical modeling; scientific computing.

Models MITgcm; LLC4320; CESM; CMIP6.

Languages French, intermediate; Spanish, intermediate.

Publications

First-Author Publications

In progress **Hoffman, L.**, and F. Massonnet. eXplainable AI reveals sea surface temperature patterns associated with slowdowns in the decline of Arctic September sea ice extent in CESM2-LE.

2025 **Hoffman, L.**, F. Massonnet, and A. Sticker. Probabilistic forecasts of September Arctic sea ice extent at the interannual timescale with data-driven statistical models. *Journal of Geophysical Research: Machine Learning and Computation*, 2, e2025JH000669. DOI: 10.1029/2025JH000669

2025 **Hoffman, L.**, M. Mazloff, S.T. Gille, D. Giglio, and P. Heimbach. Evaluating the trustworthiness of explainable artificial intelligence (XAI) methods applied to regression predictions of Arctic sea-ice motion. *Artificial Intelligence for the Earth Systems*, 4, e240027. DOI: 10.1175/AIES-D-24-0027.1

2023 **Hoffman, L.**, M. Mazloff, S.T. Gille, D. Giglio, P. Heimbach, C. Bitz, and K. Matsuyoshi. Machine learning for daily forecasts of Arctic sea-ice motion: an attribution assessment of model predictive skill. *Artificial Intelligence for the Earth Systems*, 2, 230004. DOI: 10.1175/AIES-D-23-0004.1

2022 **Hoffman, L.**, M. Mazloff, S.T. Gille, D. Giglio, and A. Varadarajan. Ocean salinity response to atmospheric river precipitation events in the California Current System. *Journal of Physical Oceanography*, 52, 1867–1885. DOI: 10.1175/JPO-D-21-0272.1

2018 **Hoffman, L.**, and T.T. Ngo. Affordable solar thermal water heating solution for rural Dominican Republic. *Renewable Energy*, 115, 1220–1230. DOI: 10.1016/j.renene.2017.09.046

Co-Author Publications

2026 Dong, X., F. Massonnet, Y. Nie, B. Richaud, Y. Gao, **L. Hoffman**, Y. Wang, and Q. Yang. Incorporating subsurface oceanic variables improves seasonal Antarctic sea ice prediction with neural networks. *Journal of Geophysical Research: Atmospheres*, 131, e2025JD045470. DOI: 10.1029/2025JD045470

2025 Giglio, D., J. Sala, J. Gilson, **L. Hoffman**, and B. Kawzenuk. Adaptive sampling of the upper ocean by autonomous floats during atmospheric river precipitation. *Geophysical Research Letters*, 52, e2025GL117069. DOI: 10.1029/2025GL117069

2020 Ngo, T.T., **L. Hoffman**, G. Hoople, W. Trevena, U. Shakya, and G. Barr. Surface morphology and drug loading characterization of 3D-printed methacrylate-based polymer facilitated by supercritical carbon dioxide. *The Journal of Supercritical Fluids*, 160, 104786. DOI: 10.1016/j.supflu.2020.104786

Invited Talks

2025 "Introduction to Machine Learning for Earth and Climate Scientists." International Global Atmospheric Chemistry (IGAC) Early Career Researchers Online Conference. Virtual. Oral presentation.

2025 "Machine learning is a useful tool to predict and understand sea ice motion in the Arctic." Earth Science Information Partners (ESIP) Machine Learning Cluster Meeting. Virtual. Oral presentation.

2024 "Machine learning is a useful tool to predict and understand sea-ice motion in the Arctic." Interagency Arctic Research Policy Committee (IARPC) Modelers' Community of Practice March 2024 Meeting – Combining Modeling and Machine Learning Approaches to Understanding the Arctic Earth System. Virtual. Oral presentation.

2023 "Machine learning is a useful tool to predict and understand sea-ice motion in the Arctic." AI for the Study of Environmental Risk (AI4ER) Seminar Series, University of Cambridge. Cambridge, UK.

2022 "Machine Learning for Evaluating the Drivers of Variability in Arctic Sea-Ice Dynamics to Improve Forecasting." American Meteorological Society Collective Madison Meeting, 17th Conference on Polar Meteorology and Oceanography. Madison, WI. Oral presentation.

2022 "Ocean surface salinity response to atmospheric river (AR) events in the California Current System." NOAA Science Seminar Series. Virtual. Oral presentation.

Presentations

2026 "Introduction to Machine Learning for Ocean Sciences." UC Santa Cruz Communicating Research Effectively (CORE) Workshop. Santa Cruz, CA.

2025 "AI4InSync - Optimizing Southern Ocean Observing Systems using AI." Kavli Institute for Theoretical Physics (KITP), The Physics of Changing Polar Climate. Santa Barbara, CA. Oral presentation.

2025 "Probabilistic Forecasts of Interannual Arctic Sea Ice Extent with Data-Driven Statistical Models." American Meteorological Society Denver Summit, 18th Conference on Polar Meteorology and Oceanography. Denver, CO. Oral presentation.

2025 Earth and Climate (ELIC) Seminar Series, Université catholique de Louvain. Louvain-la-Neuve, Belgium. Seminar.

2024 "Freshwater Processes in the Upper Ocean." Physical Oceanography Dissertation Symposium (PODS). Lihue, HI. Oral presentation.

2024 Workshop on the Role of Sea Ice and its Variability in the Climate System, International Centre for Theoretical Physics (ICTP). Trieste, Italy. Poster.

2024 Earth and Climate (ELIC) Seminar Series, Université catholique de Louvain. Louvain-la-Neuve, Belgium. Seminar.

2023 54th International Liège Colloquium on Ocean Dynamics: Machine learning and data analysis in oceanography. Liège, Belgium. Poster.

2023 "Machine learning is a useful tool to predict and understand sea-ice motion in the Arctic." NCAR Sea Ice Modeling Group Meeting. Boulder, CO. Seminar.

2023 "Machine learning is a useful tool to predict and understand sea-ice motion in the Arctic." Atmospheric and Ocean Sciences Forum, University of Colorado Boulder. Boulder, CO. Seminar.

2023 "Machine learning is a useful tool to predict and understand sea-ice motion in the Arctic." Scientific Machine Learning Symposium. San Diego, CA. Poster.

2023 "Machine learning is a useful surrogate model to parameterize and understand sea-ice motion in the Arctic." ECCO Annual Meeting. Pasadena, CA. Oral presentation.

2022 9th Annual FIRO Workshop. San Diego, CA. Poster.

2022 Observing, Modeling, and Understanding the Circulation of the Arctic Ocean and Sub-Arctic Seas Workshop. Seattle, WA. Poster.

2022 Center for Western Weather and Water Extremes (CW3E) Annual Meeting. San Diego, CA. Poster.

2022 "Ocean surface salinity response to atmospheric river (AR) events in the California Current System." Ocean Salinity Conference. New York, NY. Oral presentation.

2022 "Machine learning is a useful tool to predict and understand sea-ice dynamics in the Arctic." UCSD Jacobs School of Engineering Research Expo. San Diego, CA. Poster.

2022 "Machine Learning for Evaluating the Drivers of Variability in Arctic Sea-Ice Dynamics to Improve Forecasting." Ocean Sciences Meeting. Virtual. Oral presentation.

2021 "Ocean surface salinity response to atmospheric river events in the California Current System." American Meteorological Society 101st Annual Meeting. Virtual. Oral presentation.

2016 Institute of Electrical and Electronics Engineers Global Humanitarian Technology Conference. Seattle, WA. Poster.

Meetings and Workshops

2025 Antarctica InSync International Science Planning Workshop. Frascati, Italy.

2023 Mentoring Physical Oceanography Women+ to Increase Retention (MPOWIR) Pattullo Conference. Warrenton, VA.

2023 SeaSAR 2023. Svalbard, Norway.

2020 International Atmospheric Rivers Conference Sponsored Symposium. Virtual.

2020 Ocean Sciences Meeting. San Diego, CA.

Industry Experience

06/2016–06/2018 **Lead Engineer**, *Primo Wind, Inc.*, San Diego, CA

- Implemented design solutions for a small wind turbine system that improved power output and structural stability and optimized sustainability of materials and manufacturing.
- Patent: McMahon, E., and L. Hoffman (2017). High torque wind turbine blade, turbine, and associated systems and methods. U.S. Patent No. 9,797,370, U.S. Patent and Trademark Office.

Personal Interests

Bouldering, handstands, backpacking, surfing, guitar, violin, yoga, aimless wandering, and cats.